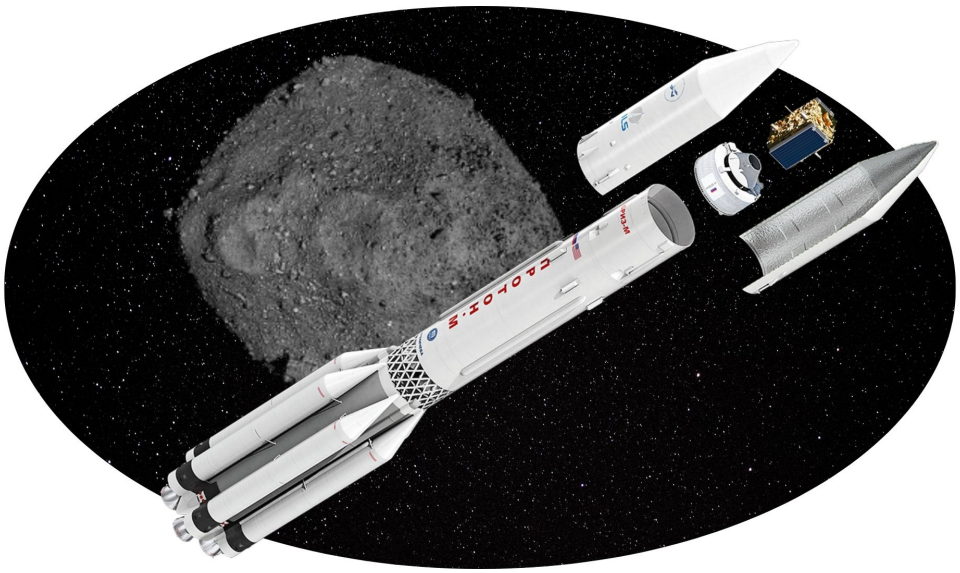
Launchers and Re-Entry 2025–2026

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Mission to Asteroid 101955 Bennu

Preliminary Mission Design & ΔV Budget Assessment



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Cover image sources: [11] and [12]

Abstract

This report presents the preliminary design of a mission to deliver a 350 kg payload to the asteroid 101955 Bennu, with the primary objective of assessing a complete ΔV budget. The mission is structured around four sequential modules: interplanetary phase design, launcher mass staging, powered ascent optimization and orbital coasting analysis. Suitable simplifying assumptions are introduced at each mission phase to enable a tractable yet representative analysis.

The interplanetary transfer, solved via Lambert's problem, identifies an optimal departure on 23 March 2030 with an arrival on 27 October 2030 (218 days), together with the total interplanetary ΔV required, also including a mid course correction. Rendezvous analysis, then confirms that the spacecraft remains within the proximity threshold for the requested time, satisfying mission requirements.

The launch vehicle is modeled as a downscaled derivative of the Proton-M with four serial stages, optimized via Lagrange-based mass redistribution, while the selected launch site is Semnan Space Centre (Iran), in order to minimize the ascent losses related to the needed inclination change.

The powered ascent is formulated as a minimum-time optimal control problem using the Pontryagin Minimum Principle, solved as a Two-Point Boundary Value problem with MATLAB's built in function `bvp4c`, aiming parking orbit insertion at prescribed altitude and inclination.

Following orbit insertion, the coasting time preceding the Trans-Orbit Insertion burn is computed.

The total required ΔV yields a positive propulsive margin, confirming mission feasibility and near optimal vehicle sizing.

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List of Acronyms

DLA Declination of the Launching Asymptote

DOF Degrees-of-Freedom

ECI Earth-Centered Inertial

EOM Equations of Motion

EUN East-Up-North

GLOW Gross Lift Off Weight

HEEJ2000 Heliocentric Earth Ecliptic J2000

ISA International Standard Atmosphere

MSL Mean Sea Level

PMP Pontryagin Minimum Principle

POI Parking Orbit Insertion

PVT Primer Vector Theory

R2BP Restricted Two-Body Problem

SOI Sphere of Influence

SRP Solar Radiation Pressure

TOF Time Of Flight

TOI Trans-Orbit Insertion

TPBVP Two-Point Boundary Value Problem

Nomenclature

A	Cross-sectional area
a	Semi-major axis
$\boldsymbol{a}_C$	Coriolis acceleration vector
$\boldsymbol{\Omega}$	Angular velocity vector
ω	Argument of the perigee
Az	Azimuth Angle, $Az = \arcsin(\cos(i)/\cos(\phi))$
β	Height scale ratio; $\beta = y/H_{scale}$
C_D	Drag coefficient
$\boldsymbol{D}$	Aerodynamic drag vector
D	Diameter
Δt	Stage burn duration
ΔV	Velocity variation
ΔV_{BVM}	Bennu Velocity Matching ΔV
ΔV_{CC}	Cruise Correction ΔV
ΔV_{req}	Required ΔV
ΔV_{TOI}	Trans-Orbit Insertion ΔV
ΔV_{avail}	Available ΔV
ΔV_{margin}	Margin ΔV
ΔV_{TOT}	Total ΔV
$\dot{\boldsymbol{y}}$	Central body dynamics vector
e	Eccentricity
Ω_E	Earth rotation rate; $\Omega_E = 7.292115$ rad/s
ϵ	Structural coefficient; $\epsilon = \frac{m_s}{m_s+m_p}$
η	Aerodynamic constant; $\eta = \frac{1}{2}\rho_0 C_D A$
f	Thrust magnitude
$\boldsymbol{g}$	Gravity vector
g_0	Earth Gravitational Acceleration at Sea Level; $g_0 = 9.80665$ km/s^2
H	Hamiltonian function
h	Altitude
H_{scale}	Atmospheric scale-height; $H_{scale} = 8440$ m
i	Inclination
I_{sp}	Specific Impulse
J	Cost function
λ	Payload ratio; $\lambda = \frac{m_{pl}}{m_s+m_p}$
G	Mass growth factor; $G = \frac{\Lambda-1}{1-\epsilon\Lambda}$
Λ	Mass ratio; $\Lambda = m_0/m_f$
ϕ	Latitude
$\dot{m}$	Mass flow rate; $\dot{m} = f/V_e$
m_p	Propellant mass
m_{pl}	Payload mass
m_s	Structural mass
m	Mass
μ	Gravitational parameter
m_w	Wet mass
n	Mean motion, $n = \sqrt{\mu/a^3}$
n_{eng}	Original launcher's number of engines
Ω	Right Ascension of the Ascending Node (RAAN)
$\boldsymbol{r}_B$	Bennu position vector (with respect to the Sun)
$\boldsymbol{r}_E$	Earth position vector (with respect to the Sun)

R_E	Earth's radius; $R_E \simeq 6371.01$ km	θ	True anomaly
ρ	Atmospheric density; $\rho = \rho_0 e^{-h/H_{scale}}$	$\boldsymbol{u}$	Control vector
ρ_0	Atmospheric density at MSL; $\rho_0 = 1.225$ kg/m^3	$\boldsymbol{\lambda}$	Costate vector
r_p	Parking orbit (Circular) radius	$\boldsymbol{v}_B$	Bennu velocity vector (with respect to the Sun)
$\boldsymbol{r}_{s/c}$	Spacecraft position vector (with respect to the Sun)	$\boldsymbol{v}_E$	Earth velocity vector (with respect to the Sun)
σ	Standard deviation	V_e	Exhaust velocity; $V_e = I_{sp} g_0$
t	Time variable	$\boldsymbol{V}_\infty$	Asymptotic escape velocity
t_a	Arrival time	$\boldsymbol{v}_{s/c}$	Spacecraft velocity vector (with respect to the Sun)
τ	Non-dimensional time variable; $\tau = t/t_f$	$v_{tr,a}$	Transfer orbit velocity at arrival
t_b	Burn-out time	$v_{tr,d}$	Transfer orbit velocity at departure
t_d	Departure time	$\boldsymbol{X}$	State vector

I. Scenario Description

The design and optimization of the mission to the asteroid 101955 Bennu are governed by a stringent set of technical requirements, centered on successfully delivering a net payload of 350 kg. A successful mission requires maintaining a relative distance of ≤ 5 km from the asteroid for a minimum continuous duration of 24 hours, necessitating active proximity manoeuvers if the natural ballistic arrival falls short of this window. To achieve this, the launcher architecture must feature a comprehensively defined staging configuration with a precise breakdown of propellant, structural, and dry masses. Furthermore, this architecture must rely entirely on flight-proven propulsion systems that strictly adhere to the engineering limits, maximum thrust capacities, and engine-count profiles of existing, historically successful launch vehicles. The mission timeline dictates that justified departure and arrival dates must be selected within the designated January 2027 to December 2036 launch window. Finally, the complete mission profile, encompassing ascent from a real-world spaceport, stage separations, Earth-departure manoeuvers, and the full interplanetary transfer, must be simulated and validated using `ode45` MATLAB's numerical integrator, maintaining absolute and relative solver tolerances of 10^{-8} or stricter across all trajectory segments.

II. Assumptions

A. Optimal mass staging

The optimal launch configuration was determined through an iterative design process coupling the Lambert-based interplanetary ΔV budget and the Bennu rendezvous requirement with a preliminary ascent allowance of 9.5–10 km/s for gravity and aerodynamic losses. Among current serial-staged architectures, which offer improved cost-efficiency ratios relative to parallel-staged configurations, the Russian **Proton-M**[*] was selected as the reference launch vehicle. Because the 350 kg payload requirement yields a Gross Lift Off Weight (GLOW) far below the nominal Proton-M capacity[†], the vehicle is modeled as a downscaled derivative of the original system, ensuring realistic thrust-to-weight ratios and structural efficiency. The two rescaling assumptions applied to achieve the final and realistic set of parameters, for the subsequent analysis, are the following:

- Structural geometry: vehicle diameter is set to 2.5 m, consistent with launchers in the 120–180 ton class[‡].
- Engine de-rating: to keep a realistic thrust-to-mass ratio, both the first and second stages are assumed to operate with a single engine, so that thrust, mass flow rate, and burn time are rescaled as:

$$f_{\text{new}} = \frac{f_{\text{total}}}{n_{\text{eng}}} \; ; \qquad \dot{m}_{\text{new}} = \frac{\dot{m}_{\text{total}}}{n_{\text{eng}}} \; ; \qquad t_{b,\text{new}} = t_{b,\text{old}} \cdot n_{\text{eng}} \tag{1}$$

[*]Nominal parameters available at Kyle [1].
[†]See Table 4.
[‡]Some examples are: Angara 1.1, CZ-6, India SSLV, Jielong, KSLV-2, Lijian 1. Further information at Kyle [2].

B. Environmental and orbital modeling

The overarching assumption throughout all mission phases is that the launcher and spacecraft are modeled as point masses. The dynamics are therefore described by a 3-DOF translational model, neglecting attitude and rotational effects in favor of trajectory optimization. The propulsion system is assumed to be fully throttleable and restartable, allowing engines to be switched on and off as required. Phase-specific assumptions are detailed in the following sections.

Ascent phase modeling:

- Local flat Earth and homogeneous gravity field: Variations in $\boldsymbol{g}$ are neglected, consistent with altitudes (300 km) much smaller than Earth's radius.
- Launch site at Mean Sea Level (MSL): Launch site's altitude is negligible[§].
- Exponential atmospheric model: $\rho(h) = \rho_0 e^{-h/H_{scale}}$: Simpler than the International Standard Atmosphere (ISA) model; it captures the essential drag variations for a preliminary ascent analysis.
- Thrust, mass flow rate and specific impulse assumed piecewise constant: Thrust and $\dot{m}$ are constant per stage; altitude-dependent pressure effects on I_{sp} are neglected.
- Topocentric EUN reference frame centered in the launch site (non-inertial): Accounts for Earth's rotation via Coriolis acceleration; centrifugal acceleration is neglected as a lower-order effect.
- Simplified aerodynamics: constant $C_D = 0.5$, zero lift: C_D is independent of Mach/Reynolds numbers; lift is assumed to be zero (no dependence with angle of attack).
- Problem statement: minimize the time required to achieve a prescribed set of orbital velocity and altitude.
- Optimal control (PMP and PVT): Thrust vector follows $\boldsymbol{u} = -\boldsymbol{\lambda_V} / ||\boldsymbol{\lambda_V}||$.
- Fixed pre-defined staging times: Staging occurs at the burnout times of the specific stage configuration.
- Multi-stage continuity: Mass makes discrete jumps when staging happens; position, velocity, and costates remain continuous throughout.

Parking phase modeling:

- Circular parking orbit: Stable 300 km altitude ($r_p = R_E$ + 300 km), with no consideration of Earth's surface altitude variations.
- Impulsive injection at perigee: Instantaneous burn at perigee of departure hyperbola for maximum energy efficiency.
- In-plane motion: Neglected out-of-plane steering; focus on the primary high-thrust energy requirements.

Interplanetary phase modeling:

- Optimal site selection and no safety/geopolitical limits: Launch site coordinates are treated as an optimization variable; no range safety or geopolitical constraints are enforced[¶].

[§] In `main_ascent_tf.m`, it is possible to account for the actual launch site's altitude by acting on the initial y-position (line 85).

[¶] Launch sites with missing or unavailable dataset entries are discarded from the candidate pool prior to analysis.

- Patched conics approximation: Simplifies the mission into distinct gravity fields with instantaneous Sphere of Influence (SOI) transitions.
- Impulsive ΔV: Neglects finite-burn duration; models manoeuvers as instantaneous velocity vector changes.
- Restricted Two-Body Problem (R2BP): Only the Sun's gravity is considered during transfer, neglecting planets and third-body effects.
- Neglected non-Keplerian forces: Solar Radiation Pressure (SRP), planetary oblateness, and third-body perturbations are omitted for simplicity.
- Point-mass center-to-center Lambert formulation: both bodies are modeled as point masses defined only by their position and velocity vectors. The transfer is computed from Earth's center to the target's center, neglecting the parking orbit altitude due to its negligible scale relative to interplanetary distances.
- High-Fidelity JPL data in HEEJ2000 for planetary ephemeris: Planetary states are obtained from JPL Horizons (Solar System Dynamics Group, NASA JPL [3]) with a 24-hour sampling interval, assumed to provide an adequate trade-off between ephemeris accuracy and computational cost. These data are then rotated in the Earth-Centered Inertial (ECI) reference frame.

Arrival and rendezvous phases:

- Negligible gravity: Bennu's gravity is neglected relative to the Sun's gravity due to the asteroid's small mass and dimensions.
- Instantaneous velocity matching: The rendezvous manoeuver is modeled as an impulsive ΔV to match the target's velocity and position at the predicted arrival time.

III. Implementation

The entire mission architecture was implemented and simulated within the MATLAB environment. The computational framework incorporates real-world constraints by integrating high-fidelity planetary ephemerides, active launch site coordinates, and current operational launch vehicle specifications featuring serial staging configurations. The design and optimization process was structured into four sequentially integrated modules: interplanetary trajectory and rendezvous analysis, launcher's choice and mass staging optimization, ascent phase analysis and orbital coasting time analysis. This modular architecture ensures a rigorous, logical data flow, wherein the output parameters of each preceding section directly seed the inputs of the subsequent phase.

A. Interplanetary phase and rendezvous analysis – `main_transfer.m`

The workspace is initialized by importing the JPL ephemeris data for Earth and Bennu and the database of available launch sites via `load_horizons_data.m` and `load_sites_data_xlsx.m`, respectively. For each departure–arrival date pair (t_d, t_a) in the Jan 2027 - Dec 2036 window, the interplanetary transfer arc is computed within `lambert_loop.m`

by solving Lambert's problem through the `Lambert.m` solver from Oldenhuis [4]:

$$\boldsymbol{v}_{tr,d},\, \boldsymbol{v}_{tr,a} = f_{\texttt{Lambert}}(\boldsymbol{r}_E(t_d),\, \boldsymbol{r}_B(t_a),\, \text{TOF},\, \mu_\odot) \tag{2}$$

The resulting hyperbolic excess velocities are used to evaluate the launch energy, the Trans-Orbit Insertion (TOI), and the Bennu rendezvous ΔV; a fixed mid-course correction margin of 0.1 km/s is added to obtain the interplanetary phase cost[‖], visualized via porkchop plots generated by `plot_porkchop.m`. The optimal launch site is then selected by `launch_site.m` from the T.S. Kelso [6] database based on the equatorial excess velocity vector $\boldsymbol{v}_{\infty,\text{eq}}$: candidate sites are filtered by Declination of the Launching Asymptote (DLA) ($DLA = \arcsin(v_{z_{\infty,eq}}/||\boldsymbol{v}_{\infty,eq}||)$) and compatible latitude, sorted by proximity of their launch azimuth to the eastward direction ($Az \to 90°$), finally the optimal site parameters are returned. The departure hyperbola and parking orbits are subsequently reconstructed by `orbit_parameters_Curtis.m` from Curtis [7], which derives the perigee state vector $\boldsymbol{X} = (\boldsymbol{r}, \boldsymbol{v})$ and the associated Keplerian elements from the equatorial excess velocity. Finally, `rendezvous.m` initializes the spacecraft heliocentric state at t_d as

$$\left[\boldsymbol{r}_{s/c}(0),\, \boldsymbol{v}_{s/c}(0)\right]^T = \left[\boldsymbol{r}_E(t_d),\, \boldsymbol{v}_E(t_d) + \boldsymbol{v}_\infty\right]^T \tag{3}$$

and propagates the trajectory via `ode45` under the two-body Equations of Motion (EOM) $\dot{\boldsymbol{y}} = [\boldsymbol{v},\, -(\mu/r^3)\boldsymbol{r}]^T$ in two stages: from departure to the nominal arrival epoch; over an additional 50-day post-rendezvous interval, following the impulsive arrival manoeuver, from which the continuous time spent within the 5 km proximity threshold is assessed.

B. Mass staging - `main_staging.m`

The performance of the multi-stage Proton M launch vehicle is assessed through an ideal staging reconstruction followed by a Lagrange-based mass optimization procedure using the analytical expressions given in Sagliano [5].

The function `ideal_mass_staging.m` first reconstructs the baseline launcher configuration starting from the stage structural masses, propellant masses, specific impulses, and payload mass. For each stage, the corresponding mass ratio, structural coefficient, and velocity contribution are evaluated through the Tsiolkovsky rocket equation.

Subsequently, the function `optimal_mass_staging.m` redistributes the required mission ΔV among the stages in order to minimize the total lift-off mass. The optimization is performed by solving (with `fsolve`) for the parameter λ^*, according to the constraint:

$$\sum_{i=1}^{n} V_{e,i} \ln \underbrace{\left(\frac{1 - \lambda^*/V_{e,i}}{\epsilon_i}\right)}_{\Lambda_i^*} - \Delta V_{req} = 0 \tag{4}$$

The resulting optimal mass ratios Λ_i^* are then converted back into stage masses through the mass growth factor G_i:

$$G_i = \frac{\Lambda_i^* - 1}{1 - \epsilon_i \Lambda_i^*} \tag{5}$$

[‖] The value of $\Delta V_{CC} = 0.1$ km/s is taken by Sagliano [5].

This allows the determination of the updated propellant masses, structural masses, wet masses, and burnout masses for the complete launcher configuration. Finally, the propulsion performance parameters, including propellant mass flow rate and stage burn times, are derived from the optimized vehicle characteristics. In Table 1 the vehicle features required for the ascent analysis are summarized. These values incorporate the fundamental staging assumptions detailed in Section II.A.

Parameter	Stage 1	Stage 2	Stage 3	Stage 4	Unit
Number and Type of Engines*	1 × RD-276	1 × RD-0210	1 × RD-0213	1 × S5.98M	-
D*	2.5	2.5	2.5	2.5	m
m_p	118280	25562	4487.8	716.68	kg
m_s	8450.8	1787.6	337.34	85.785	kg
m_0	160061	33327	5977.6	1152.5	kg
m_f	41778	7765.2	1489.80	435.78	kg
$\dot{m}$*	590.7173	181.1927	191.44	6.10	kg/s
f in vacuum*	1830600	581040	613896.29	19613.30	N
I_{sp} in vacuum	316	327	327	328	s
t_b*	200.2345	141.0764	23.44	117.54	s

* Parameters derived from the mass, diameter rescaling and engine de-rating process.

Table 1 Technical specifications and optimized design parameters for the downscaled Proton-M configuration.

C. Optimal ascent - `main_ascent_tf.m`

The ascent phase is formulated as a three-dimensional minimum-time optimal control problem based on Pontryagin Minimum Principle (PMP) and Primer Vector Theory (PVT), stated as a TPBVP. The implementation extends the two-dimensional flat-Earth framework of Longuski et al. [8] with the following modifications: (i) a third spatial dimension to accommodate out-of-plane steering for inclination change; (ii) Coriolis acceleration in the rotating topocentric frame; (iii) multi-stage events with discrete mass jumps at burnout; (iv) modified state and costate equations in `ascent_odes_tf_3stage.m` and boundary conditions in `ascent_bcs_tf_3stage.m`. The state vector $\boldsymbol{X}$ includes position and velocity components, while the control vector $\boldsymbol{u}$ defines the thrust direction. The objective is to minimize the final time required to reach a 300 km circular parking orbit:

$$\min J = \int_0^{t_f} 1 = t_f \tag{6}$$

The dynamics are formulated in a non-inertial topocentric reference frame and include thrust, aerodynamic drag, gravity, and Coriolis acceleration. The EOM are given by:

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{V} \\ \dot{\boldsymbol{V}} = \frac{f(t)}{m(t)}\boldsymbol{u} - \frac{1}{m(t)}\boldsymbol{D} + \boldsymbol{g} + \boldsymbol{a}_C \end{cases} \tag{7}$$

where the drag force and Coriolis acceleration are modeled as:

$$\boldsymbol{D} = \underbrace{\frac{1}{2}\rho_0 C_D A}_{\eta}\ e^{-\overbrace{y/H_{scale}}^{\beta}}\|\boldsymbol{V}\|\boldsymbol{V}\ ; \quad \boldsymbol{a}_C = -2\boldsymbol{\Omega}\times\boldsymbol{V} \quad \text{with:} \quad \boldsymbol{\Omega} = \Omega_E\ [\ 0,\ \sin(\phi),\ \cos(\phi)\ ]^T \tag{8}$$

The optimal control structure $\boldsymbol{u}^*$ is derived from the Hamiltonian (in Mayer form):

$$H(\boldsymbol{X},\boldsymbol{u},\boldsymbol{\lambda},t) = \boldsymbol{\lambda}^T\boldsymbol{f}(\boldsymbol{X},\boldsymbol{u},t) \ \xrightarrow{\text{PMP}}\ \boldsymbol{u}^* = \arg\min_{||\boldsymbol{u}||=1} H \ \xRightarrow{\text{PVT}}\ \boldsymbol{u}^* = -\frac{\boldsymbol{\lambda}_V}{\|\boldsymbol{\lambda}_V\|} \ \text{ for } \boldsymbol{\lambda}_V \neq 0 \tag{9}$$

Costate dynamics is defined by $\dot{\boldsymbol{\lambda}} = -\nabla_{\boldsymbol{X}} H$ and the resulting TPBVP is solved numerically using MATLAB `bvp4c`, where the boundary conditions are defined at launch pad and circular orbit insertion at 300 km altitude with a prescribed inclination. The numerical implementation adopts a non-dimensional formulation and a normalized time variable $\tau = t/t_f \in [0,1]$. The solution is obtained through a five-step continuation scheme, gradually introducing mass variation, aerodynamic drag, and staging events. To verify the optimality and physical feasibility of the generated outputs, the script performs a post-processing analysis:

- Calculation of the total required ΔV via numerical integration (with `trapz`) of:

$$\Delta V = \int_{t_i}^{t_f} \frac{f(t)}{m(t)}\,dt \tag{10}$$

  This solution is cross-validated against the theoretical Tsiolkovsky rocket equation.
- Forward propagation using `ode45` to verify terminal constraints;
- Hamiltonian consistency check ($H \approx 0$) along the trajectory and comparison with the one obtained via propagation.

D. Coasting analysis - `parkingOrbitVisualizer`

The transition from ascent to interplanetary injection is evaluated through geometric timing constraints and numerical propagation in the inertial frame. The implementation uses the analytical expressions given in Vallado [9] and Lasagni Manghi [10]. To avoid costly plane-change manoeuvers, launch must occur during a valid alignment window. After Parking Orbit Insertion (POI), the spacecraft coasts until the departure geometry is satisfied. The coasting time is computed from the orbital mean motion (n). The trajectory is validated through numerical propagation using `ode45` of the EOM in `gravity_dynamics.m`. The motion is integrated in two phases: coasting up to the hyperbolic perigee and post-impulsive escape after the TOI manoeuver.

E. Interplanetary trajectory visualization - `interplanetaryVisualizer`

To generate a complete propagated trajectory from Earth to Bennu, the interplanetary trajectory is computed and plotted by integrating the EOM with `ode45` from the departure to the arrival dates. The resulting plot depicts the orbital positions of Earth and Bennu throughout the entire temporal window (Jan 2027 – Dec 2036), whereas the spacecraft's position is indicated exclusively between the departure and arrival epochs.

IV. Numerical Results

A. Interplanetary phase and rendezvous

1. Lambert problem

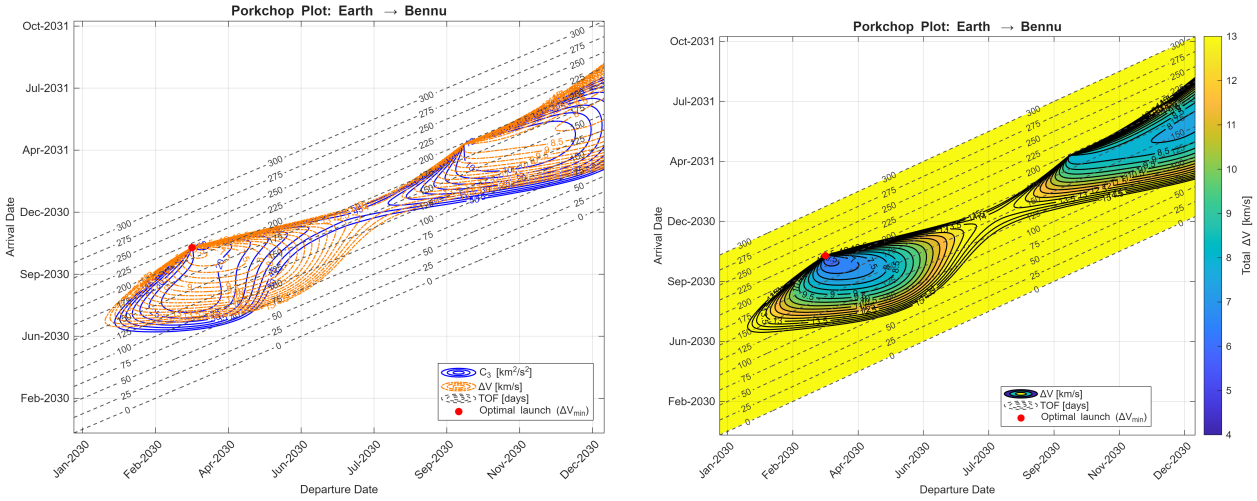
By solving the Lambert problem, the key interplanetary trajectory parameters are determined. These results are summarized in Table 2:

Parameter	Value	Unit
Departure Epoch	23 March 2030	-
Arrival Epoch	27 October 2030	-
TOF	218	days
Departure C_3	20.63	km^2/s^2
Departure $\boldsymbol{V}_\infty$	$[-3.4897,\ 0.0473,\ -2.9064]^T$	km/s
ΔV_{TOI}	4.1077	km/s
Arrival $\boldsymbol{V}_\infty$	$[-1.4212,\ 1.1322,\ 0.3579]^T$	km/s
ΔV_{BVM}	1.8519	km/s
Total ΔV (ΔV_{TOT})*	6.0597	km/s

*__Note__: The total ΔV includes a minor $\Delta V_{CC} = 0.1$ km/s mid-course cruise correction.

Table 2 Interplanetary trajectory parameters from Lambert's problem solution.

The corresponding porkchop plots are presented in Figure 1.



(a) Porkchop plot with ΔV and C_3 isolines. **(b) Porkchop plot with ΔV isolines.**

Fig. 1 Porkchop plots in the reduced 2030 visualization.

2. Parking orbit and departure hyperbola

By taking as input the components of the hyperbolic excess velocity and propagating back the trajectory it's possible to define the hyperbolic transfer and, consequently, the parking orbit parameters are presented in Table 3:

Parameter	Parking orbit	Departure hyperbola	Unit
a	6671.01	−19323.91	km
e	0	1.345220	-
i	35.66	35.66	deg
Ω	251.03	251.03	deg
ω	0	131.98	deg
θ	131.98	0	deg

Table 3 Orbital parameters of the parking orbit and the departure hyperbola.

3. Optimal Launch Site

Based on the previously defined parking orbit requirements, Semnan Space Centre (Iran), located at [35.23°N, 53.92°E], is selected as the optimal launch site. Its geographic latitude of 35.23°N closely matches the required parking orbit inclination of 35.66° (Table 3), thereby minimizing the out-of-plane ΔV associated with inclination changes. The corresponding launch geometry yields a launch azimuth of 84.08°, confirming a predominantly eastward trajectory consistent with energy-optimal ascent.

4. Rendezvous

According to the proximity operations analysis, the spacecraft remains within the 5 km range for more than 80 hours, no additional station-keeping or trajectory correction manoeuvers are required. This compliant behavior is illustrated in Figure 2.

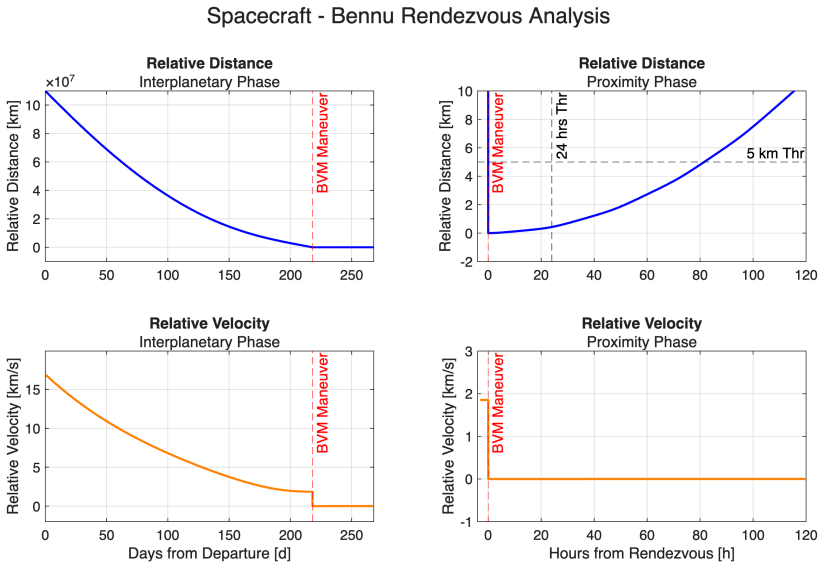


Fig. 2 The left plots show the macro-scale interplanetary approach as a function of days from departure. The right plots display the close-proximity rendezvous phase in hours, illustrating the relative distance remaining below the 5 km operational threshold.

B. Mass staging

For a given payload of 350 kg, the optimization process for the Russian Proton-M vehicle, yielded the design parameters summarized in Table 4.

Parameter	Stage 1	Stage 1 Opt	Stage 2	Stage 2 Opt	Stage 3	Stage 3 Opt	Stage 4	Stage 4 Opt	Unit
m_s	30600	8450.8	11000	1787.6	3500	337.34	237	85.785	kg
m_p	4.283×10^5	1.1828×10^5	1.573×10^5	25562	46562	4487.8	19800	716.68	kg
m_w	4.589×10^5	1.2673×10^5	1.683×10^5	27350	50062	4825.2	22170	802.47	kg
m_0	-	1.6006×10^5	-	33327	-	5977.6	-	1152.5	kg
ϵ	0.066681	0.066681	0.065359	0.065359	0.069913	0.069913	0.1069	0.1069	-
Λ	2.5776	3.8312	2.882	4.2919	2.7895	4.0123	8.2794	2.6446	-
ΔV	2934.3	4162.4	3394.3	4671.4	3289.7	4455.4	6799.1	3128.2	m/s

Table 4 Mass staging results and performance values: comparison of baseline stage parameters against optimized values derived from the optimal mass staging algorithm.

The total ΔV achievable is: $\Delta V_{avail} = 1.64174 \times 10^4$ m/s.

To evaluate the impact of the optimization process, Figure 3 illustrates a comparative analysis of the propellant allocation.

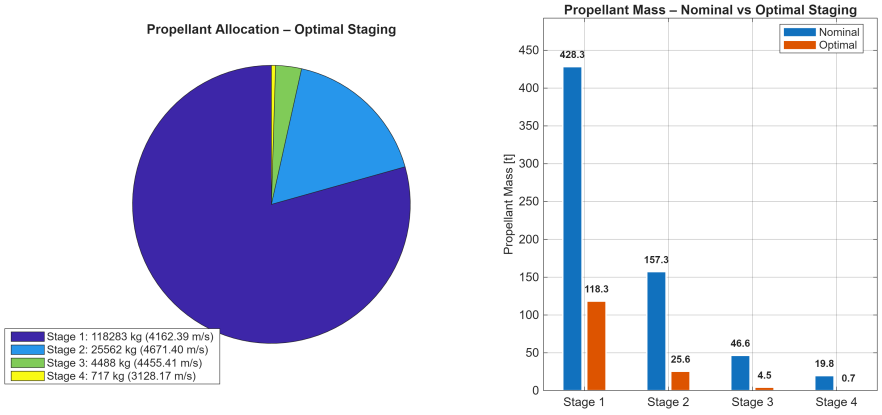


Fig. 3 *Left*: breakdown of optimal propellant mass allocation and associated ΔV increments per stage. *Right*: comparison between nominal and optimized propellant mass configurations across all stages.

C. Ascent phase and coasting time

1. Optimal ascent

The results of the ascent trajectory analysis are summarized in Tables 5–8, which report the stage-wise propulsion and mass characteristics, the comparison between numerical and analytical ΔV estimates and the statistical dispersion of the final state variables. During ascent, the first two stages are fully expended and part of the third stage is consumed to achieve parking orbit insertion; the residual ΔV of the third stage and the entire fourth stage are reserved for the interplanetary manoeuvers and are therefore not included in Tables 5 and 6.

Parameter	Stage 1	Stage 2	Stage 3	Unit
m_0	1.6006×10^5	33327.0	5977.6	kg
m_f at t_b	41778.0	7765.2	3868.2	kg
Δt	200.2344	141.08	11.019	s
Absolute Staging Time	200.2344	341.3108	352.3297	s
h	97.93	297.06	300.00	km
f	1.8306×10^6	5.8105×10^5	6.139×10^5	N

Table 5 Multi-stage ascent performance and mass characteristics.

Stage	Numerical ΔV	Ideal ΔV	Margin	Unit
Stage 1	4.1302	4.1624	0.0322	km/s
Stage 2	4.6386	4.6714	0.0328	km/s
Stage 3	1.2532	1.3957	0.1425	km/s
Total Ascent ΔV	**10.2171**	**10.2295**	**0.0124**	km/s

Table 6 Numerical vs. ideal analytical ΔV comparison.

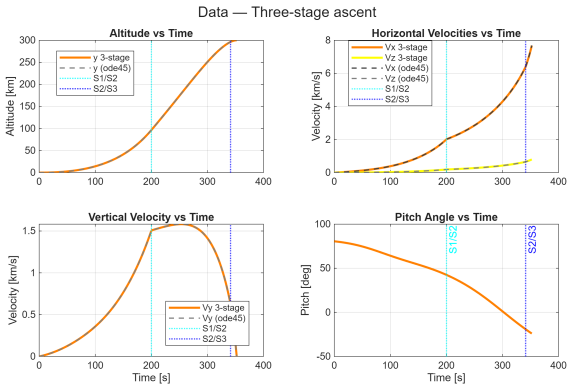
State Variable	Propagated Value	Target Value	Unit
y_f	300.0000	300.0000	km
V_{x_f}	7.684622	7.684623	km/s
V_{y_f}	0.000000	0.000000	km/s
V_{z_f}	0.796191	0.796191	km/s

Table 7 Terminal state comparison at parking orbit injection: comparison between `ode45` forward-propagated values and `bvp4c` target boundary conditions.

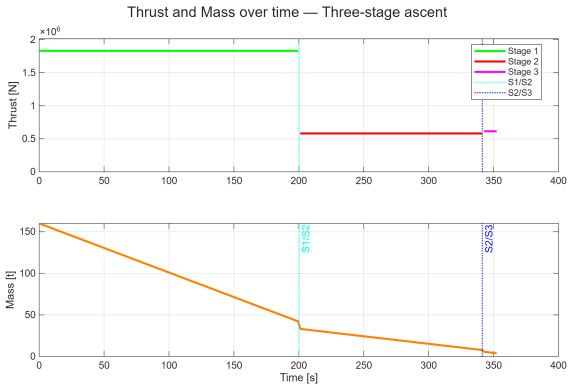
Parameter	Value	Unit
σ_x	0.5344	m
σ_y	0.6586	m
σ_z	0.0585	m
σ_{V_x}	1.7717	m/s
σ_{V_y}	1.3321	m/s
σ_{V_z}	0.1801	m/s
σ_H	0.000107	-

Table 8 Standard deviations of position, velocity, and Hamiltonian variables.

Figures 4–7 summarize the results obtained from the optimal ascent trajectory simulation. In particular, the reported quantities describe the vehicle state evolution, thrust and mass profiles, costate dynamics, optimal control behavior, Hamiltonian evolution, trajectory reconstruction, and final injection accuracy. The vertical dashed lines highlight the stage separation instants.

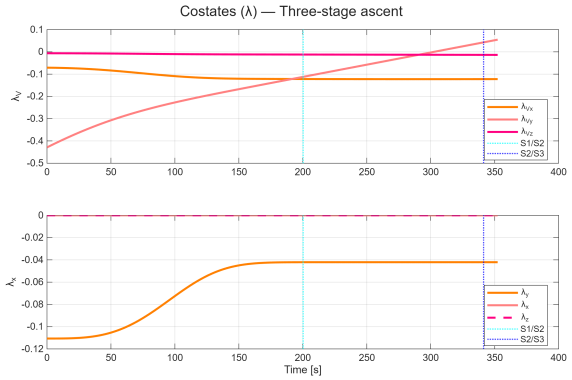


(a) Evolution of states and pitch angle.

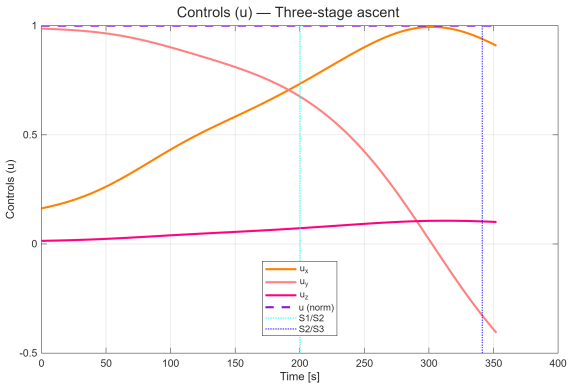


(b) *Up*: Thrust evolution. *Down*: Mass evolution.

Fig. 4 Time histories of the main ascent states, thrust profile, and vehicle mass evolution. The discrete mass drops at the stage separation instants and the thrust reduction are clearly visible, reflecting the sequential expulsion of each stage structure.

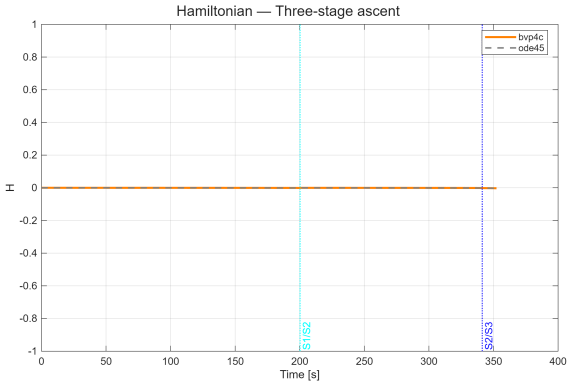


(a) Position and velocity costates along the ascent.



(b) Optimal control vector components along the ascent.

Fig. 5 Evolution of the costate variables and optimal control vector components during the ascent phase. The first derivatives vary along the trajectory due to the presence of atmospheric drag; however, they are constant when drag is neglected, in agreement with theoretical expectations. Both costates and control vector components are continuous, confirming the feasibility of the solution.



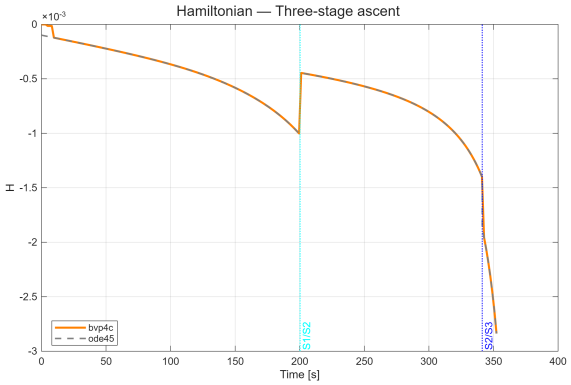
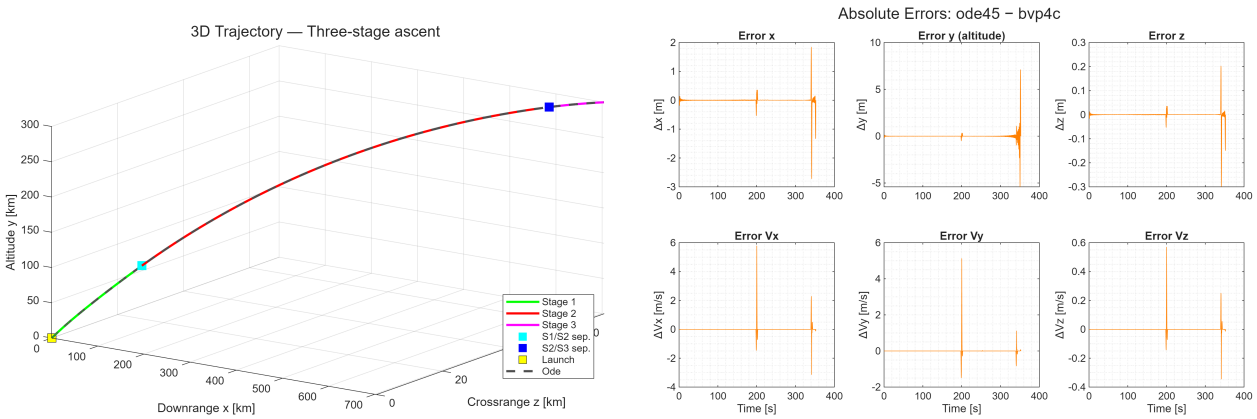


Fig. 6 Hamiltonian evolution during the optimal ascent trajectory. H is not strictly constant because the ratio $f(t)/m(t)$ introduces explicit time dependence (via continuous mass depletion and discrete staging events) making the system non-autonomous ($\partial H/\partial t \neq 0$). Nevertheless, $H \approx 0$ is maintained throughout.



(a) 3D representation of the optimized ascent trajectory. **(b) Residual errors along the trajectory.**

Fig. 7 Three-dimensional comparison between the ascent trajectory computed with `bvp4c` and the propagated solution, along with the residual errors evaluated along the trajectory.

2. Coasting phase

Following parking orbit insertion, the spacecraft must coast until the perigee of the departure hyperbola aligns with the required TOI burn point. The ascent phase covers an arc of 6.06°, after which a coasting arc of 44.23° is required before the TOI manoeuver can be executed. From the parking orbit mean motion, this translates into a waiting time of approximately 11.10 minutes.

V. Conclusions - Final ΔV budget assessment

As presented in Table 9, the mission ΔV budget analysis yields a total required ΔV of 16.28 km/s (when numerical values for the ascent are considered), compared to an available capability of 16.42 km/s, resulting in a positive propulsive margin of +0.141 km/s. When ideal ascent ΔV values are used consistently throughout the budget, the corresponding margin reduces to +0.128 km/s. Both values confirm compliance with mission requirements while indicating a near-optimal launch vehicle sizing. Table 10 provides a breakdown of the ΔV budget by mission phase and stage.

Parameter	ΔV **[km/s]**
ΔV_{avail}	16.4174
ΔV_{req} (Num)	16.2768
ΔV_{req} (Ideal)	16.2892
ΔV_{margin} (Num)	**+0.1405**
ΔV_{margin} (Ideal)	**+0.1282**

Table 9 Summary of available ΔV capabilities, total mission requirements, and resulting propulsive margins.

Phase	**Stage**	ΔV **ideal [km/s]**	**Stage Status**
Ascent	Stage 1	4.1624	Fully expended
Ascent	Stage 2	4.6714	Fully expended
Ascent	Stage 3 (partial)	1.3957	Partially expended
ΔV_{TOI}	Stage 3 (residual)	3.0597	Fully expended
ΔV_{TOI}	Stage 4 (partial)	1.0480	Partially expended
ΔV_{CC}	Stage 4 (partial)	0.1000	Partially expended
ΔV_{BVM}	Stage 4 (partial)	1.8519	Partially expended
Margin	Stage 4 (residual)	**+0.1282**	-

Table 10 Mission ΔV budget breakdown by phase and stage.

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